INSTALLING THE PROVISION BARRIERS:

If straps are installed but one or both breakers will not be at this time, provision barriers MUST be installed. (Skip this section if two breakers are installed.)

23. **For three-phase systems: Refer to Figure 9**
Form two barriers (Item 7) into the shape of a "U", invert one over the other, and slide over the B-phase strap end. Secure the two barriers onto the B-phase strap with a 3/8"-16 screw (Item 17).

24. **For both single and three-phase systems: Refer to Figures 9 and 10.**
Form two barriers (Item 7) into the shape of a "U", then invert the "U" and slide over each of the A- and C-phase strap ends. Secure each with a 3/8"-16 screw (Item 17).

If second breaker is NOT installed, repeat steps 23 & 24 for the other provision location on the straps.

INSTALLING THE NEUTRAL LUG:

25. **Refer to Figure 11a and 11b**
If this kit is installed in a panelboard, install the Panelboard Neutral Lug Assembly (Item 10).
If this kit is installed in a switchboard, install the Switchboard Neutral Lug Assembly (Item 11).

INSTALLING COVER PLATE AND FILLERS:

26. Reassemble the deadfront assembly to the interior (removed in Step 3).

27. **Refer to Figure 12**
Install the 6.25" cover plate (Item 4) by aligning it over the 6.25" of selected unit space and related breaker(s). Use four 1/4"-20 x 3/8" screws (two from Item 15 and two from when the cover was removed in step 2) to secure the cover plate to the deadfront rails. Torque all hardware as specified on the rear of the deadfront. Mark the circuit ID cards (Item 8) as desired and mount them to the cover plate by placing each behind a clear card holder (Item 9) and pushing the two into the proper pair of holes in the cover plate next to the proper circuit. Repeat for each location. Finally, use blank fillers (Item 5) at every location that doesn't have a breaker penetrating through the cover plate, using bolt (Item 13) and nut (Item 14) in opposing corners.

HANDLE EXTENSIONS:

Handle extension recommended for ease of breaker operation. One handle extension ships per breaker. Refer to instruction 11-B-1088-01 for more details.

28. If breaker handle will be exposed, a handle extension (ships with the breaker) can be mounted to the breaker handle for ease of use.

29. If breaker handle will end up behind a cover, and handle extension can not be mounted, place it in the clear pocket for easy access.

COMPLETING THE INSTALLATION:

30. **Ensure that any bus or breaker hardware is torqued as specified on the rear of the deadfront. Then replace all the cover plates and gutter covers. Finally, ensure any fillers are in place BEFORE re-energizing the equipment.**

Figure 9

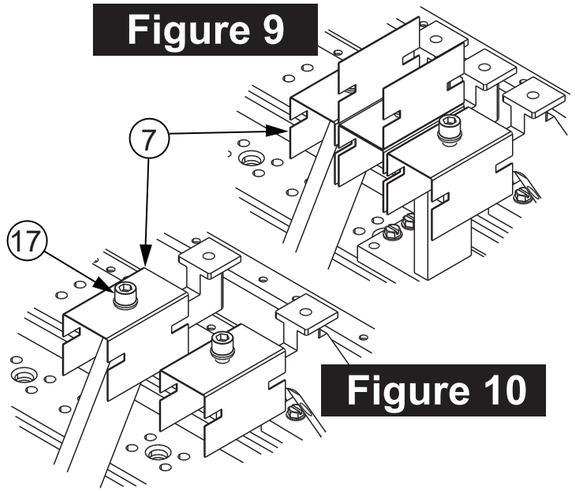


Figure 10

Figure 11b

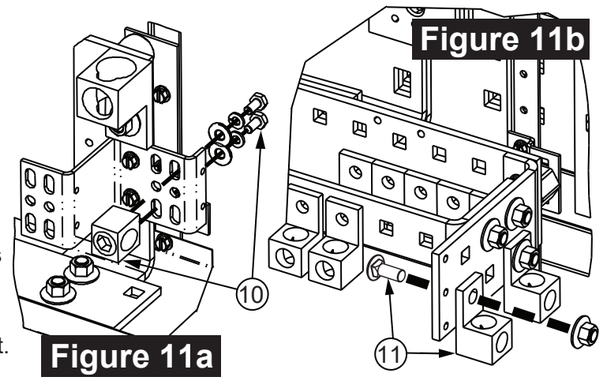
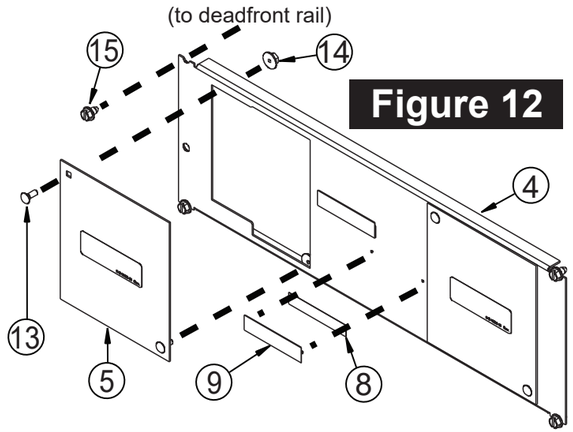


Figure 11a

Figure 12



These instructions do not purport to cover all details or variations in equipment, or to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise, which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the local **Siemens** sales office. The contents of this instruction manual shall not become part of or modify any prior or existing agreement, commitment or relationship. The sales contract contains the entire obligation of **Siemens**. The warranty contained in the contract between the parties is the sole warranty of **Siemens**. Any statements contained herein do not create new warranties or modify the existing warranty.

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SIEMENS

Distribution Switchboard / P5 38" wide Power Panel Mounting Kit S3VA53T2D / S3VA54T2D for Twin-mount 3VA53, 3VA63, 3VA54 or 3VA64 breakers

Installation Instructions

Installation Instructions

The following instructions are for the installation of Siemens strap kit **S3VA53T2D** or **S3VA54T2D** in a Distribution Switchboard or a 38" wide P5 Power Panel. Kit **S3VA53T2D** is for use with a **Siemens Type 3VA53 (MJAS, HJAS or CJAS)** or **Type 3VA63** (**MJAE**, **HJAE**, **CJAE** or **LJAE**) breaker as a branch device in a **400A** application. Kit **S3VA54T2D** is for use with a **Siemens Type 3VA53 (MJAS, HJAS or CJAS)** or **Type 3VA63** (**MJAE**, **HJAE**, **CJAE** or **LJAE**) or **Type 3VA54** (**MLAS**, **HLAS** or **CLAS**) or **Type 3VA64** (**MLAE**, **HLAE** or **CLAE**) breaker as a branch device in a **600A** application. The parts provided in this kit connect **Type 3VA53, 3VA63, 3VA54** or **3VA64** breakers to a 1-phase, 2 or 3-wire system or a 3-phase 3 or 4-wire system. This kit requires 6.25" of unit space.
NOTE: The breakers and breaker lugs are NOT included in this kit and must be purchased separately.

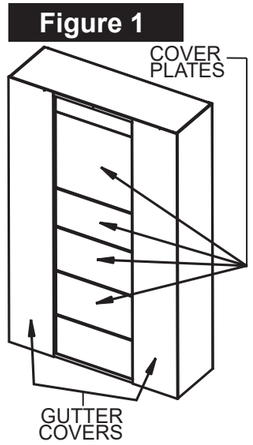
THIS KIT CONTAINS THE FOLLOWING ITEMS:

ITEM	DESCRIPTION	QTY	TORQUE
1	A/C-phase Strap *	2	N/A
2	B-phase Strap *	1	N/A
3	Breaker Mounting Bracket	2	N/A
4	6.25" Cover Plate	1	N/A
5	Filler Plate	2	N/A
6	Interphase Barrier	1	N/A
7	Provision Barrier	8	N/A
8	Circuit ID Card	2	N/A
9	Card Holder	2	N/A
10	Panelboard Neutral Branch Lug Assy.	**	N/A
11	Switchboard Neutral Branch Lug Assy.	**	N/A
12	#10-32 Breaker Mounting Screw	4	20 lb-in
13	#10-24 x 1/2" Carriage Bolt	4	N/A
14	#10 Lock Nut	4	20 lb-in
15	1/4"-20 x 3/8" Hex Head Washer Screw	6	72 lb-in
16	1/4"-20 x 1/2" Hex Head Washer Screw	4	72 lb-in
17	3/8"-16 x 1-1/8" SEMS Screw	6	240 lb-in
18	Bus Hardware Kit, Strap to Section ***	1	72 lb-in

* Components different for 400A vs. 600A applications.
** Application dictates amount of these components used.
*** This sub-kit of parts is packaged in a bag labeled "Standard Bus Hardware Kit". It includes an instruction card and Qty. 18 of fasteners (three different lengths). Select the five required per the instruction card and discard the rest.



Figure 1



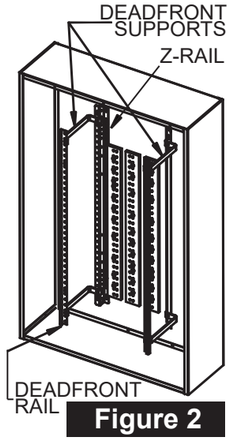


Figure 2

PREPARING THE EQUIPMENT:

1. **Lock off power supplying this equipment before working on it.**
2. **Refer to Figure 1**
   Remove the two gutter covers and cover plates.
3. **Refer to Figure 2**
   This kit requires 6.25" of empty unit space. To locate the proper mounting position, measure from the top deadfront support to the intended location of the empty unit space (filler) on the deadfront. Transfer this dimension from the deadfront support along the Z-rail and mark. This will be the unit space (as shown in Figures 3 & 4). After marking the Z-rail, remove the deadfront rails.
4. If an existing branch module occupies the same location chosen for this kit, remove all of its devices, components and parts.

INSTALLING THE MOUNTING BRACKETS:

5. **Refer to Figure 3**
   Orient one of the mounting brackets (Item 3) as shown and attach it to the Z-rail with two 1/4"-20 screws (Item 16).
6. Repeat for the second mounting bracket.
7. Tighten the four screws to the torque specified on the rear of the deadfront.

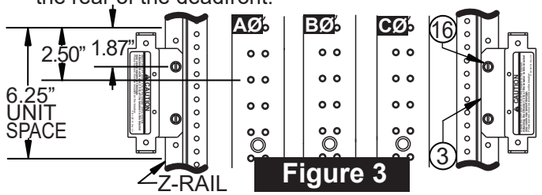


Figure 3

INSTALLING THE BREAKER STRAPS:

8. **Refer to Figure 4**
   For three-phase installation ONLY:
   (Skip to Step 10 for single-phase systems)
   To install the B-phase strap (Item 2), locate it as shown, and attach it to the B-phase bus with four 1/4"-20 screws (Item 18) thru the strap and into the bus.
9. Tighten the four screws to the torque specified on the rear of the deadfront.

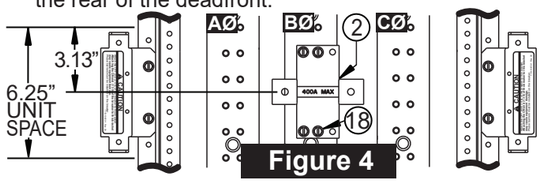


Figure 4

10. **Refer to Figure 5**
    To install the A and C-phase straps (Item 1), locate them as shown, and attach to the A and C-phase bus with four 1/4"-20 screws (Item 18) each, thru the straps and into the bus.
11. Tighten the screws to the torque specified on the rear of the deadfront.

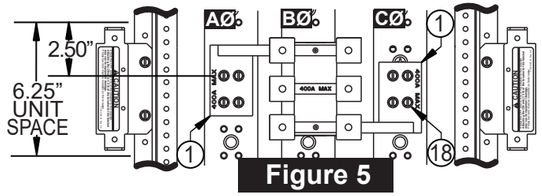


Figure 5

NOTE: For provisions, skip to the INSTALLING THE INTERPHASE BARRIER section

PREPARING THE SIEMENS BREAKER:

For Single Barrel lugs: Refer to Figure 6 and the separate instructions that ship with the lugs.

a. Remove the four screws that secure the end cover and slide the cover off.
b. Insert the nutkeepers into the breaker end until they click into place.
c. Remove and set aside the set screw from each of the single barrel lugs.
d. Slide each single barrel lug into the breaker end openings.
e. Insert the lug mounting screws thru the set screw opening and tighten to **142 lb-in** to secure each single barrel lug to their respective nutkeeper.
f. Re-insert the set screws removed in Step c.
g. Slide the end cover back onto the breaker end and re-secure with the four screws removed in Step a.

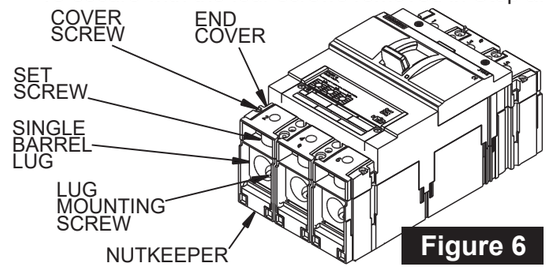


Figure 6

For Double Barrel lugs: Refer to Figure 7 and the separate instructions that ship with the lugs.

a. Insert the nutkeepers into the breaker end until they click into place.
b. Place a washer and mounting screw into each lug and slide into position, tightening the screw to **355 lb-in** to secure it to the nutkeeper.

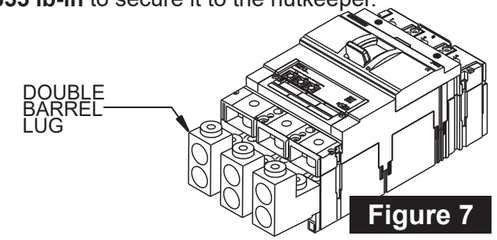


Figure 7

NOTE: A 400A breaker can be installed in a 600A twin kit or kit rated for 600A.

INSTALLING THE SIEMENS BREAKER:

12. **Refer to Figure 8**
    Orient the breaker so that the breaker stab end faces the inside and rests on the straps, the lug end faces the gutter and rests on a breaker mounting bracket, and the breaker rests between the two breaker mounting bracket fins.
13. If mounting with Double Barrel lugs, then slide plastic bottom extension piece under the lug end. (If Single Barrel lugs, skip to the next step.)
14. Secure the breaker to the breaker mounting bracket using two #10-32 breaker mounting screws (Item 12). Do not tighten at this time.
15. Remove the end cover of the breaker strap end by removing the four screws and sliding the cover off. (This allows easy access to secure breaker stabs to the phase straps.)
16. Attach the breaker on the strap end with 3/8"-16 screws (Item 17) into the straps, one per phase.
17. Tighten the screws to the torque specified on the rear of the deadfront.
18. Replace and re-secure the end cover after properly tightening to the straps.
19. If mounting with Double Barrel lugs, then slide plastic top extension piece over the lug end. Secure with four screws. (If Single Barrel lugs, skip to the next step.)
20. Follow the same steps for the other breaker.

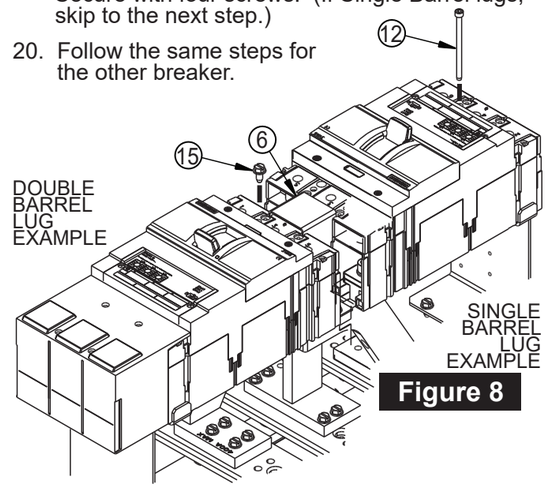


Figure 8

INSTALLING THE INTERPHASE BARRIER:

NOTE: Interphase barrier MUST be installed in all cases, even when breakers are NOT present.

21. **Refer to Figure 8**
    Slide the interphase barrier (Item 6) over and through the slots on either side of the B-phase straps between the two breakers, as shown.
22. Insert a 1/4"-20 x 3/8" screw (Item 15) on each side of the barrier into the outer straps and tighten.

(more instructions on following page)